Logic circuits

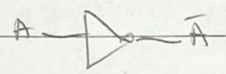
Logic gates and inverters

Basic properties of logic gates

- A logic gate should have the following properties:
  - ↳ The binary o/p gives the req. logic function of the i/p(s)
  - ↳ Quantisation of amplitudes is req. → non-linear devices
  - ↳ Signal amplitudes should be regenerated in passing through the gate → significant gain
  - ↳ Capable of accepting more than one i/p (no. of independent i/ps is the fan-in)
  - ↳ Capable of driving more than one o/p (no. of independent o/ps driven is the fan-out)
  - ↳ Changes at the o/p of the gate do not affect the signals at the i/p

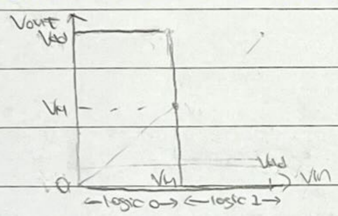
Operation and basic properties of inverter

- An inverter is the simplest logic gate w/ one i/p and one o/p

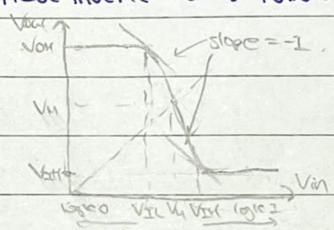


A	$\bar{A}$
0	1
1	0

- The transfer characteristics of an ideal and typical inverter is as follows



ideal inverter



typical inverter

↳ $V_{in}=0$, $V_{out}=V_{dd}$; $V_{in}=V_{dd}$, $V_{out}=0$	↳ i/p and o/p voltages deviate from ideal values
↳ infinitely sharp transition at $V_{in}=V_M$	↳ gradual transitions at $V_{in}=V_M$.
↳ zero propagation delay	↳ finite propagation delay.

- V_{IH}: min. voltage recognised as logic 1 ; V_{IL}: max. voltage recognised as logic 0.

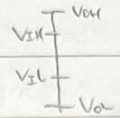
  V_{OH}: min. o/p voltage when o/p is HIGH ; V_{OL}: max. o/p voltage when o/p is LOW.
- The inverter switching threshold V_M is defined as the pt the voltage transfer curve intersects w/ the line $V_{out}=V_{in}$ → represents the pt. at which the inverter switches state. (usually $V_M=\frac{V_{dd}}{2}$, but not always)

Designing "real" logic gates

- The i/p and o/p signals must be consistent
  - ↳ If two gates of the same logic family are connected, $V_{OH} > V_{IH}$ for HIGH to be recognised as logic 1 and $V_{OL} < V_{IL}$ for LOW to be recognised as logic 0
- V_{IL}, V_{IH}, V_{OL}, V_{OH} characterise the behaviour of a logic family
- We only req. $V_{in} \leq V_{IL} - \varepsilon$ or $V_{in} \geq V_{IH} + \varepsilon$ to be recognised if it weren't for the presence of noise.

Noise and noise margins

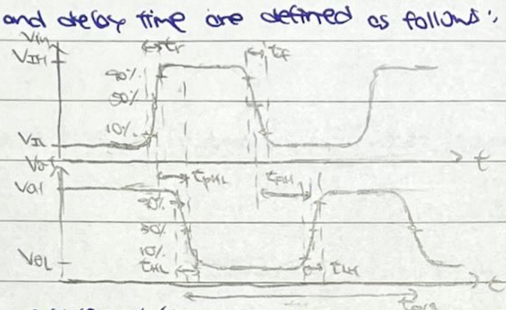
- Noise is spurious signals generated in a system mainly due to inductive or capacitive coupling btwn signal lines.
- If the noise amplitude is large enough, it may change the signal enough to change its logic level.
- Noise margin is a measure of how much noise can be added to the logic i/ps of a circuit w/o the circuit responding improperly, and depend on the transfer characteristics
  ↳ Low state noise margin : $\boxed{NM_L = V_{IL} - V_{OL}}$
  ↳ High state noise margin : $\boxed{NM_H = V_{OH} - V_{IH}}$



- For the gates to work correctly and w/o spurious o/ps, we req. $\boxed{NM_L > 0 \text{ and } NM_H > 0}$.
- The logic swing is the expected range of signals at the o/p., $\boxed{\text{logic swing} = V_{OH} - V_{OL}}$

Rise time, fall time and delay time

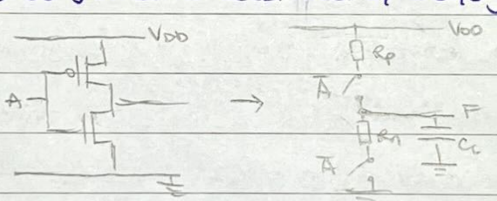
- The rise time, fall time and delay time are defined as follows:



t_r: rise time (i/p) ; t_f: fall time (i/p) ; t_{HL}: fall time (o/p) ; t_{LH}: rise time (o/p)

t_{PHL}: propagation time high-low ; t_{PLH}: propagation time low-high ; t_{cyc}: cycle time.

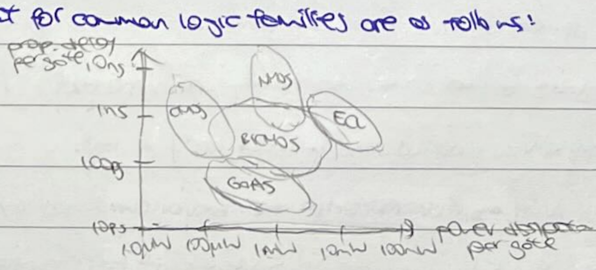
- Propagation delays of CMOS gates can be determined by analysing RC equivalent circuits



For a pure RC circuit, rise time = 2.2RC, fall time = 2.2RC.

Power-delay product

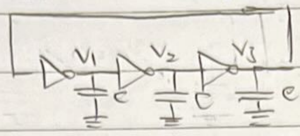
- The power-delay product is the product of power consumption (averaged over a switching event) times the i/p-o/p delay.
- The power-delay product for common logic families are as follows:

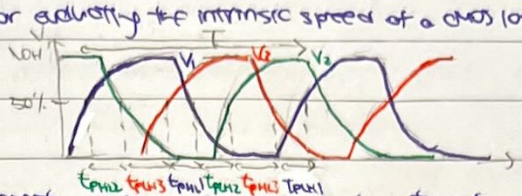


Application of inverters

① Ring oscillator

- A ring oscillator is formed by connected an odd no. n of inverters connected in chain.
- The ring oscillator is a fundamental circuit for evaluating the intrinsic speed of a CMOS logic process.





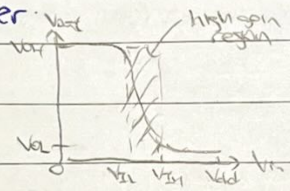
Assuming the time delay for each inverter is const, i.e. $t_{pHL1} = t_{pHL2} = t_{pHL3} = t_{pHL}$, $t_{pLH1} = t_{pLH2} = t_{pLH3} = t_{pLH}$

$$\boxed{f = \frac{1}{T} = \frac{1}{n(t_{pHL}+t_{pLH})}}$$

* we can also define the propagation delay t_p to be $\boxed{t_p = \frac{1}{2}(t_{pHL}+t_{pLH})}$

② Amplifier

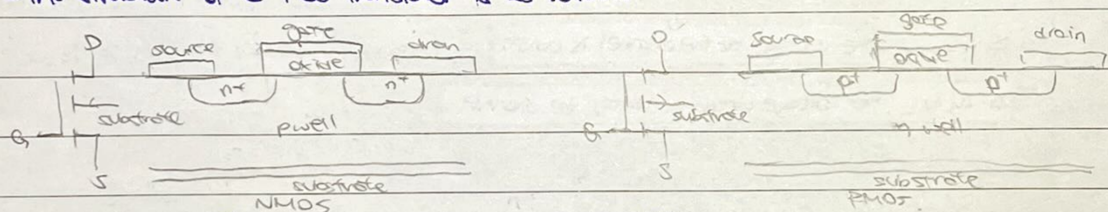
- For Vin btwn V_{IL} and V_{IH}, the inverter gain > 1. → inverter can act as linear amplifier.
- Applying a resistive load to the inverter and operating the inverter in this region is equivalent to a common source amplifier.



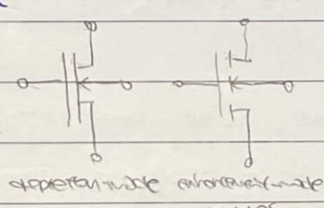
MOS transistor and logic circuits

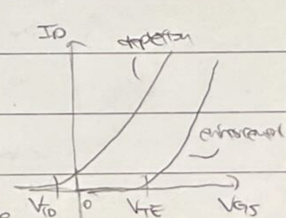
MOS transistor

- The structure of a MOS transistor is as follows:



- The NMOS/PMOS contains the following key elements:
  - ↳ Si substrate containing p-type/n-type impurities
  - ↳ Two heavily doped n^+/p^+ regions (source, drain regions).
  - ↳ Inversion layer under gate to connect source and drain. (depends on gate voltage)
  - ↳ Gate oxide is a thin film of insulating SiO_2.
- The flow of current btwn source and drain is controlled by the gate voltage. (Gate voltage modulates the conductivity of the inversion layer).
- MOSFETs can be depletion mode or enhancement mode
  - ↳ Depletion mode: $V_{T,D} < 0$ → normally ON
  - ↳ Enhancement mode: $V_{T,E} > 0$ → normally OFF





MOS transistor operation (Assume $V_B = V_S = 0$)

① Accumulation ($V_{GS} < 0$, any V_{DS})

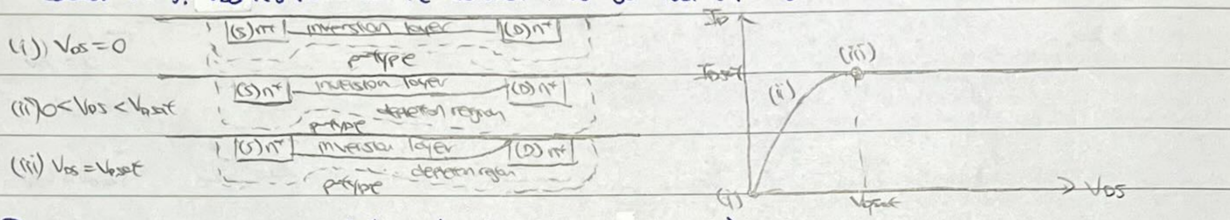
- Source to drain path consists of two reverse-biased diodes → no current

② Depletion / cut-off ($V_{GS} < V_T$, any V_{DS})

- Electrostatic field attracts e^- towards the gate, but there is still a deficit of e^- (so channel is highly resistive) → no drain current can flow.

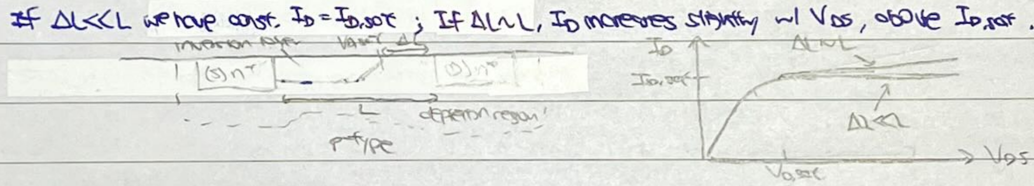
③ Inversion, linear/triode ($V_{GS} > V_T$, $0 \leq V_{DS} \leq V_{D,sat} = V_{GS} - V_T$)

- An inversion layer is included in the channel and connects source and drain. The induced channel acts like a resistor → current depends linearly on drain voltage V_D.
- As V_{DS} increases, the depletion region widens and e^- conc. in the inversion layer decreases on the drain side → channel conductance decreases → I_D-V_D slope decreases.
- The inversion layer eventually vanishes on the drain side → pinch off, I_D-V_D curve is flat. (saturation). V_{DS} has reached the saturation voltage V_{Dsat} and I_D is the saturation current $I_{D,sat}$.



④ Inversion, saturation ($V_{GS} > V_T$, $V_{DS} \geq V_{D,sat} = V_{GS} - V_T$)

- As V_{DS} increases above V_{Dsat}, the channel length L effectively changes by ΔL. The region of the channel ΔL is depleted → highly resistive → voltage increases in $V_{DS} > V_{D,sat}$ are dropped across this part.
- If $\Delta L \ll L$ we have const. $I_D = I_{D,sat}$; If $\Delta L \sim L$, I_D increases slightly w/ V_{DS}, above $I_{D,sat}$.



I_D-V_D characteristics.

- For n-type MOSFET, when there is a continuous channel, i.e. $V_{GS} > V_T$, $0 \leq V_{DS} \leq V_{GS} - V_T$ [linear]

$$\boxed{I_D = \frac{k}{2}\left(2(V_{GS} - V_T)V_{DS} - V_{DS}^2\right)}$$

for very small V_{DS}, we approx. $I_D = k(V_{GS} - V_T)V_{DS} \rightarrow R_D = \frac{1}{k(V_{GS} - V_T)}$

For n-type MOSFET, when the channel is pinched-off, i.e. $V_{GS} > V_T$, $V_{DS} \geq V_{GS} - V_T$ [saturation]

$$\boxed{I_D = \frac{k}{2}(V_{GS} - V_T)^2}$$

- Ideally, I_D is independent of V_{DS} in the saturation region, but increasing V_{DS} causes an electronic shortening of the channel → increase in drain current (Early effect).

$$\boxed{I_D = \frac{k}{2}(V_{GS} - V_T)^2(1 + \lambda V_{DS})}$$

where λ is the channel length modulation coefficient. ($V_A = \frac{1}{\lambda}$ is the Early voltage).

- k is the transconductance parameter

Transconductance parameter k.

- The transconductance parameter k depends on factors related to physical and electrical characteristics of the device materials, and dimensions of the conductive channel, length L, width W.

$$\boxed{k = k'\left(\frac{W}{L}\right) = \mu C_{ox}\left(\frac{W}{L}\right) = \mu\frac{\varepsilon_{ox}}{t_{ox}}\left(\frac{W}{L}\right)}$$

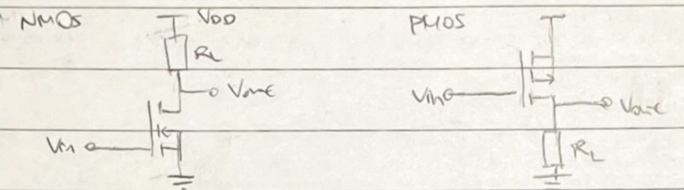
where ε_{ox} is the permittivity of the SiO_2 gate insulator, | determined from material characteristics.

t_{ox} is the thickness of the SiO_2 gate insulator.

μ is the mobility of the charge carrier ← obtained empirically from electrical measurements

NMOS and PMOS inverter

- The circuit diagram for the NMOS inverter and the PMOS inverter is as follows:



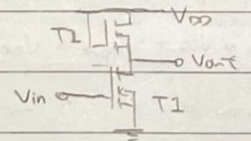
The resistor R_L forms a load. The p.d. across R_L is determined by the current driven through it by the MOSFET (the driver).

- Desireable values of R_L would be
  - ↳ A high value to reduce power consumption and bring V_{OL} towards $0/V_{DD}$ for NMOS/PMOS inverter.
  - ↳ A low value to ensure V_{OH} approaches $V_{DD}/0$ (NMOS/PMOS) rapidly when the o/p terminal is loaded w/ a capacitance. → reduce t_{PLH}/t_{PHL} (NMOS/PMOS).
  - → the req. are in conflict so any choice is a compromise.
- Voltage transfer characteristic can be gained by plotting I_D vs V_{DS} for the resistor and the transistor then finding intersections for various V_{GS}. Alternatively, directly equate the load and driver currents.

general strategy — assume the regime the transistor is operating if unknown and check if the answer is contradictory.

NMOS inverter w/ NMOS load

- Load resistors are much larger than transistors → not economical to fabricate → use transistor as load.

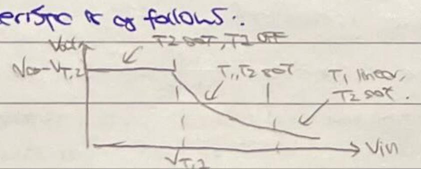


saturation gives poor resistive characteristics.

Transistor behave like a pseudo-resistor w/ the gate connected to the drain (always in saturation)

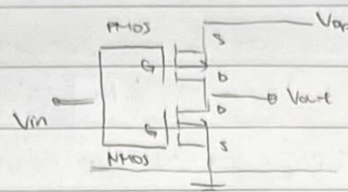
- A much smaller $\frac{W}{L}$ ratio (smaller k) for the load device to enhance its resistive properties → ratioed logic.
- o/p in HIGH state → load must conduct, i.e. $V_{GS,2} \geq V_{T,2}$; $V_{out} = V_{DD} - V_{DS,2} = V_{DD} - V_{GS,2} \leq V_{DD} - V_{T,2}$ → V_{OH} reduced.

  o/p in LOW state → $V_{GS,2}$ is max → resistance of load min. → opposite of what we want (poor performance)
- The typical transfer characteristic is as follows:

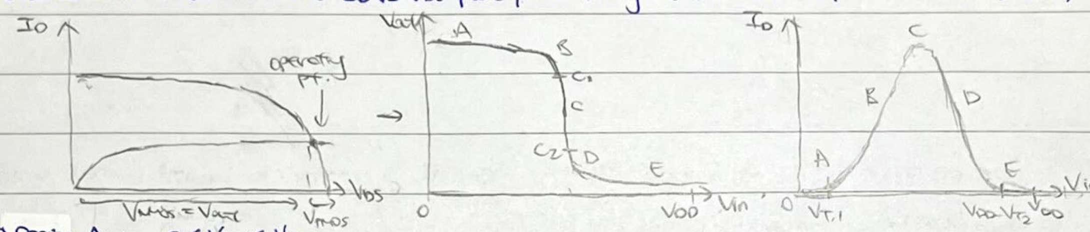


CMOS inverter

- The circuit diagram for the CMOS device is as follows



- The o/p is pulled to one of the power rails by a conductive (ON) device. The other device serves as a load of effectively infinite resistance → static properties closer to ideal inverter.
- The challenge is that PMOS devices are rel. slow ($\mu_p < \mu_n$) → device needs to be physically larger to compensate, → more complex fabrication process.
- The voltage transfer characteristic can be found by plotting the NMOS and flipped PMOS characteristics on the same I_D vs V_{DS} plot, and finding the intersection pts (for various V_{GS})



(i) Region A : $0 \leq V_{in} < V_{T,N}$

↳ NMOS OFF, PMOS ON (linear)

↳ No current flows (since NMOS is OFF).

(ii) Region B : $V_{TN} \leq V_{in} < V_M$

↳ NMOS ON (saturated), PMOS ON (linear)

↳ NMOS starts to conduct

(iii) Region C : $V_{in} = V_M = V_{out}$

↳ NMOS ON (saturated), PMOS ON (saturated)

$I_{DN} = I_{DP} \rightarrow K_N (V_{in} - V_{TN})^2 = K_P ((V_{DD} - V_{in}) - |V_{TP}|)^2 \rightarrow \boxed{V_M = \dfrac{V_{DD} - |V_{TP}| + V_{TN}\sqrt{K_N/K_P}}{1 + \sqrt{K_N/K_P}}}$

↳ If $K_P = K_N$ and $V_{TN} = |V_{TP}|$, then $\boxed{V_M = \frac{V_{DD}}{2}}$

↳ V_{out} at pts C_1 and C_2 can be found using the V_{in} above and

C_1: $|V_{DSP}| = |V_{GSP}| - |V_{TP}| = V_{DD} - V_{in} - |V_{TP}| \rightarrow V_{out} = V_{DD} - |V_{DSP}| = V_{in} + |V_{TP}|$

C_2: $V_{DSN} = V_{GSN} - V_{TN} = V_{in} - V_{TN} \rightarrow V_{out} = V_{DSN} = V_{in} - V_{TN}$

(iv) Region D : $V_M < V_{in} \leq V_{DD} - |V_{TP}|$

↳ NMOS ON (linear), PMOS ON (saturated)

(v) Region E : $V_{DD} - |V_{TP}| \leq V_{in} \leq V_{DD}$

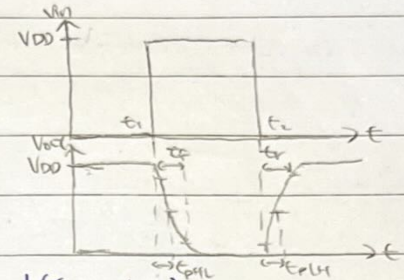
↳ NMOS ON (linear), PMOS OFF

↳ No current flows (since PMOS is OFF)

* V_{IH} and V_{IL} can be found from $\frac{dV_{out}}{dV_{in}} = -1$ → we find $\boxed{NM_H = NM_L \approx 0.4 V_{DD}}$
* Alternatively, the voltage transfer characteristic / current draw can be found by projecting the line of intersection of current surfaces $I_{DN} = f_1(V_{in}, V_{out})$, $I_{DP} = f_2(V_{in}, V_{out})$

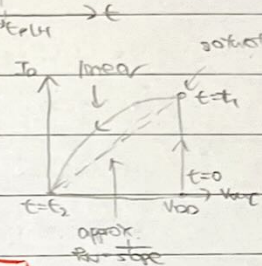
CMOS inverter switching speed

- Assume the i/p signal is ideal — MOSFETs snap in and out conduction instantaneously.
- Propagation delay arises from the time taken for load capacitance C_L to be charged/discharged to the switching pt. $V_M = V_{DD}/2$.



- We define the average propagation delay t_P as $t_P = \frac{1}{2}(t_{PHL} + t_{PLH})$
- Consider the average channel resistance R_N

$$R_N = \frac{V_{DD}}{\frac{1}{2}K_N(V_{DD}-V_{TN})^2}$$



It can be shown that

$$\boxed{t_{PHL} \approx 0.7 R_N C_L \approx \frac{C_L}{K_N V_{DD}}} \qquad \boxed{t_{PLH} \approx 0.7 R_P C_L \approx \frac{C_L}{K_P V_{DD}}}$$

Thus the average propagation delay is $t_P = \frac{1}{2}(t_{PHL} + t_{PLH}) = \frac{C_L}{2V_{DD}}\left(\frac{1}{K_N} + \frac{1}{K_P}\right)$

- Similar calculations yield the rise time t_r and fall time t_f

$$\boxed{t_f \approx 0.5\frac{C_L}{K_N}} \qquad \boxed{t_r \approx 0.5\frac{C_L}{K_P}}$$

where $V_{DD} \sim 5V$, $V_{TN} = |V_{TP}| = 1V$.

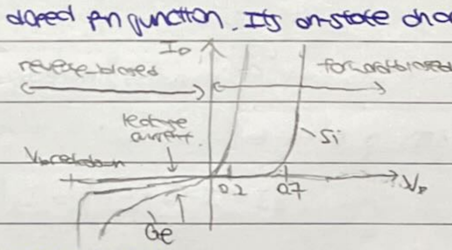
* capacitor discharge through NMOS (so t_{PHL}, t_f depend on K_N)
capacitor charge through PMOS (so t_{PLH}, t_r depend on K_P)

- For gates other than NOT, we may have two NMOS/PMOS → series: $\left(\frac{W}{L}\right)_{eff} = \frac{W}{2L}$, parallel: $\left(\frac{W}{L}\right)_{eff} = \frac{2W}{L}$

Bipolar transistors and logic circuits

Diodes.

- The diode is formed by a doped p-n junction. Its on-state characteristics are



The current of an ideal diode is given by

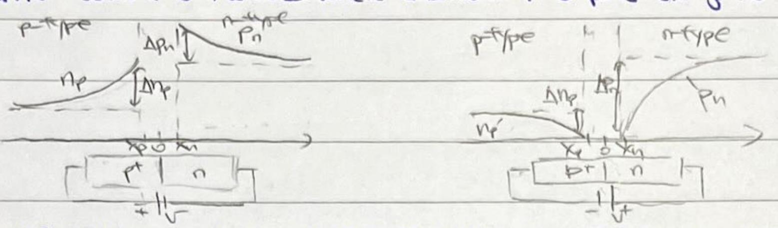
$$\boxed{I_D = I_S\left[e^{\frac{V_D}{V_t}} - 1\right] \approx I_S e^{\frac{V_D}{V_t}}}$$

where $\boxed{V_t = \frac{kT}{q}}$ is the thermal voltage ($V_t = 25mV$ at RT)

and I_S is the reverse saturation current. ($I_S \sim 10^{-15} - 10^{-14}$ A for IC diodes)

Transient behaviour of diodes

- when the diode is forward-biased, minority carriers are injected from other side of the junction. Current is carried by majority carriers.
- when the diode is reverse-biased, only minority carriers contribute to current flow (leakage current). Minority carrier current is increased where additional h^+-e^- pairs are generated (e.g. heat, light).

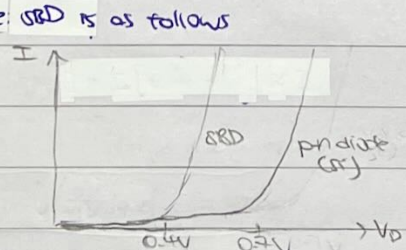


- The time to switch btwn conductivity and non-conductivity is determined by the time taken to establish minority carrier distributions.
  - ↳ Turn ON: diode switch into conduction rel. quickly — time taken determined by circuit capacitance
  - ↳ Turn OFF: switch-off is slow as injected minority carriers need to be removed — this can be represented by a diffusion capacitance $\boxed{C_D = A I_D = \frac{\tau}{V_t} I_D}$ τ is mean recombination time of carriers.

Schottky barrier diode (SBD)

- The SBD is a device w/ the properties of a diode created by a microscopically clean contact btwn a lightly-doped semiconductor and certain metals → unipolar device.

- A characteristic potential barrier ϕ_B of metal-semiconductor interface impedes flow of e^- from metal to semiconductor. Height of barrier depends on the materials used (typically 0.6-0.8V)
- A depletion region is set up in the semiconductor close to the surface.
- e^- in the semiconductor may acquire energy sufficient to surmount the barrier and escape to the metal → small leakage current.
- Forward bias: height of barrier reduced → e^- flow more easily from metal to semiconductor
  Reverse bias: height of barrier increased and depletion region wider.
- The I-V characteristics of the SBD is as follows



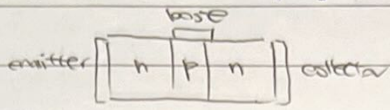
The current for SBD is similar to that of a pn junction diode

$$\boxed{I = I_0\left[e^{V/V_t} - 1\right] \approx I_0 e^{V/V_t}}$$

A SBD of a given size typically has a lower forward voltage than a comparable pn junction diode operating at the same forward current (threshold voltage V_T ~0.3V smaller)

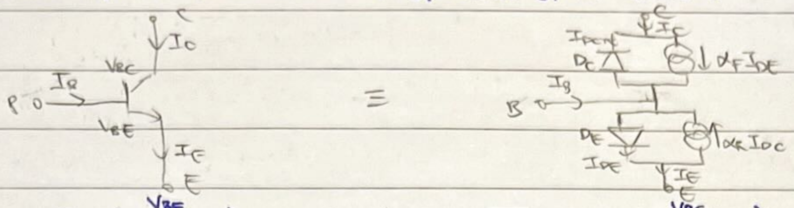
Bipolar transistors

- The bipolar transistor is formed by two back-to-back pn junctions.



- The operation of the BJT can be described mathematically using Ebers-Moll model.

(Ebers-Moll model applies to all modes of operation of the device)



$$I_{D,E} = I_{S,E}\left(e^{\frac{V_{BE}}{V_T}} - 1\right) \qquad I_{D,C} = I_{S,C}\left(e^{\frac{V_{BC}}{V_T}} - 1\right)$$

$$I_E = I_{D,E} - \alpha_R I_{D,C} \qquad I_C = \alpha_F I_{DE} - I_{DC}$$

where $I_{S,C}$, $I_{S,E}$ are the saturation currents of the collector-base, emitter-base junctions respectively

and $\alpha_F \approx 1$, $\alpha_R < 0.86$.

$$\Rightarrow \boxed{I_E = I_{S,E}\left(e^{\frac{V_{BE}}{V_T}} - 1\right) - \alpha_R I_{S,C}\left(e^{\frac{V_{BC}}{V_T}} - 1\right)} \; ; \; \boxed{I_C = \alpha_F I_{S,E}\left(e^{\frac{V_{BE}}{V_T}} - 1\right) - I_{S,C}\left(e^{\frac{V_{BC}}{V_T}} - 1\right)}$$

- For an ideal transistor, reciprocity thm. gives $\alpha_F I_{S,E} = \alpha_R I_{S,C} = I_S$

Bipolar transistor operation.

① Cut-off ($V_B < V_C$, $V_E < V_E$).

↳ Behaves like an open switch ($I_B = I_C = 0$)

↳ B-C junction, B-E junction reverse biased.

② Forward active ($V_B < V_C$, $V_B > V_E$)

↳ Behaves like an amplifier

↳ B-C junction reverse biased, B-E junction forward biased

↳ $\boxed{I_C = \beta_F I_B}$

③ saturation ($V_B > V_C$, $V_B > V_E$)

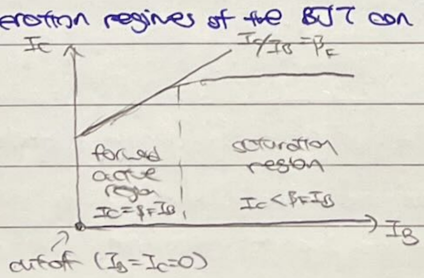
↳ Behaves like a closed switch

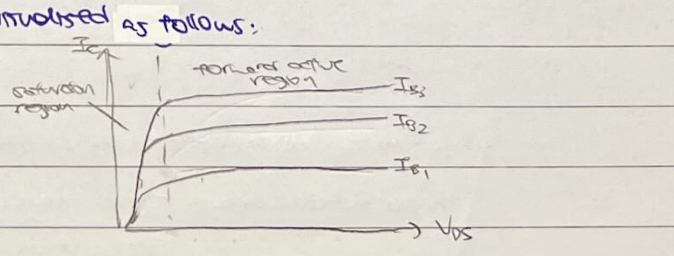
↳ B-C junction, B-E junction forward biased

↳ $\boxed{I_C < \beta_F I_B}$, Degree of saturation $\boxed{\sigma = \frac{I_C}{\beta_F I_B} < 1}$

* There is also the reverse-active region, but this operation regime is not very useful.

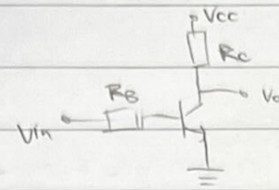
- The operation regimes of the BJT can be visualised as follows:



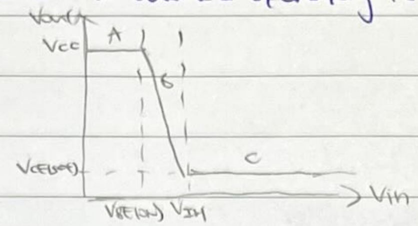


The bipolar inverter (Resistor transistor logic (RTL))

- The circuit diagram for the bipolar inverter is as follows:



- The voltage transfer characteristic can be obtained by analysing the diff operating regimes of the BJT.



(i) Region A: $0 \leq V_{in} < V_{BE(on)}$

↳ BJT OFF

↳ No current flows (since BJT is OFF)

(ii) Region B: $V_{BE(on)} \leq V_{in} < V_{BE(sat)}$

↳ BJT ON (forward active)

↳ BJT starts to conduct, $I_C = \beta_F I_B$

(iii) Region C: $V_{in} \geq V_{BE(sat)}$

↳ BJT ON (saturation)

↳ current saturates, i.e. I_C no longer increases w/ increasing I_B, $I_C < \beta_F I_B$.

↳ $V_{CE} = V_{CE(sat)}$

- V_{OH} is at cut-off region (region A) → BJT OFF → o/p pulled to V_{CC} → $V_{OH} = V_{CC}$

V_{OL} is at saturation region (region C) → $I_C = I_{C(sat)}$ → voltage drop across R_C const. → $V_{OL} = V_{CE(sat)}$

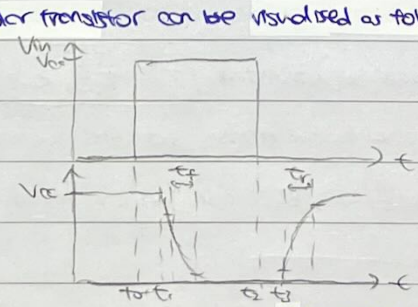
V_{IH} is at the edge of saturation → $V_{BE} = V_{BE(sat)}$, $I_C = \beta_F I_B$ → $V_{IH} = V_{BE} + I_B R_B = V_{BE(sat)} + R_B \frac{V_{CC} - V_{CE(sat)}}{\beta_F R_C}$

V_{IL} is at the edge of cut-off → $V_{BE} = V_{BE(on)}$ → $V_{IL} = V_{BE(on)}$

- As usual, $NM_H = V_{OH} - V_{IH}$, $NM_L = V_{IL} - V_{OL}$, logic swing $= V_{OH} - V_{OL}$

Bipolar inverter switching speed

- The switching delay of the bipolar transistor can be visualised as follows:



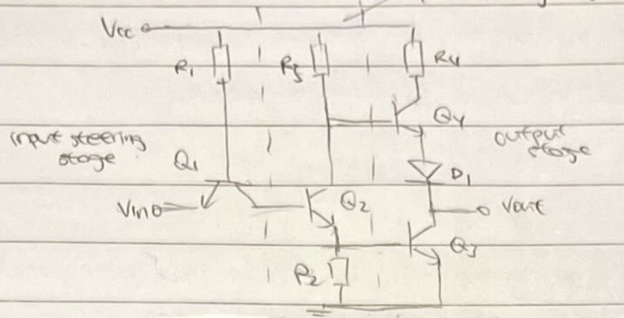
- fall time t_f: discharge of stray capacitance at o/p ; rise time t_r: charging of stray capacitance through R_C

transistor switching delays, $t_1 - t_0$: charging of stray capacitance in base circuit

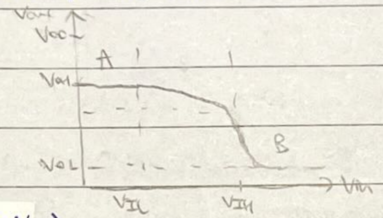
$t_3 - t_2$: discharging of stray capacitance in base circuit + removal of minority carriers from base region

Transistor-transistor logic (TTL) inverter.

- The circuit diagram for the TTL inverter is as follows: phase splitting stage



- Q_1, R_1 form the i/p steering stage:
  - ↳ Vin HIGH → B-E reverse-biased, B-C forward-biased (reverse active) → current flows to Q_2 through R_1
  - ↳ Vin LOW → B-E forward-biased, B-C reverse-biased (forward active) → current flows to i/p through R_1
- Q_2 provides extra gain + additional current to switch the o/p stage Q3 more effectively
- Q4 replaces the resistive load as an active pull-up. It has more gain and can source more current than a resistor. Q_3 and Q4 form a totem-pole o/p stage.
- The voltage transfer characteristics is as follows:



(i) Region A: Vin LOW ($V_{in} < V_{IL}$)
- ↳ Q1 ON (forward active)
- ↳ Q2 OFF → V_{B3} pulled to GND, V_{B4} pulled to Vcc → Q_3 OFF, Q4 ON (saturation)
- ↳ $V_{out} = V_{CC} - R_4 I_4 - V_{CE(sat)} - V_{D1} = V_{OH}$

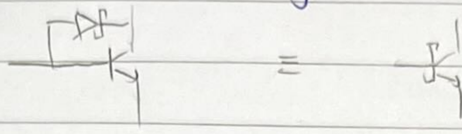
(ii) Region B: Vin HIGH ($V_{in} > V_{IH}$)
- ↳ Q1 ON (reverse active)
- ↳ Q2 ON (saturation) → Q3 ON (saturation) → $V_{E2} = V_{BE(sat)}$, $V_{C2} = V_{BE(sat)} + V_{CE(sat)}$, $V_{C3} = V_{CE(sat)}$
- ↳ $V_{B4} = V_{C2} = V_{BE(sat)} + V_{CE(sat)}$, $V_{E4} = V_{C3} + V_{D(on)} = V_{CE(sat)} + V_{BE(sat)}$ → $V_{BE4} = 0 < V_{BE(on)}$ → Q4 OFF.
- ↳ $V_{out} = V_{C3} = V_{CE(sat)} = V_{OL}$

- Typical values are $V_{BE(on)} = 0.65V$, $V_{BE(sat)} = 0.75V$, $V_{CE(sat)} = 0.2V$
  $V_{OH} = 3.6V$, $V_{OL} = 0.2V$, $V_{IH} = 1.5V$, $V_{IL} = 0.6V$ → $NM_L = 0.4V$, $NM_H = 2.1V$, logic swing = 3.4V
- TTL uses high conductance paths to drive the base → lower propagation delays.
- V_{OH} for TTL is significantly lower than Vcc, and is still relatively slow as we need to remove minority carriers when bringing the BJT out of saturation

* For gates w/ multiple i/ps, we can replace Q1 w/ a transistor w/ two emitters.

Schottky TTL (STTL)

- In a schottky transistor, a SBD is connected across the base-collector junction of each bipolar transistor node to desaturate (prevent junction from being forward biased by ~0.5V)
- Avoiding saturation allows for faster switching



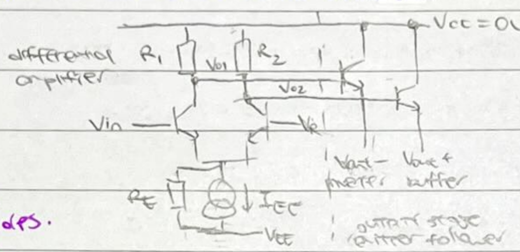
- STTL has switching times ~3ns and has twice the power consumption of TTL

delay power-consumption trade off.

Low-power STTL uses high value resistors → 10x less power consumption, 3x slower switching.

Emitter-coupled logic (ECL) inverter

- The circuit diagram for a ECL inverter is as follows:

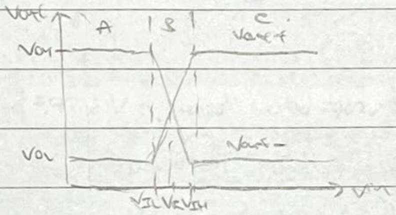


easy to get complementary o/ps.

if not given, found using $I_{EE} = \frac{V_E - V_{EE}}{R_E}$, $V_E = V_{in} - V_{BE(ON)}$

- The i/p stage is a differential amplifier w/ a constant tail current I_{EE}. Vin rel. to V_R steers I_{EE} btwn the two branches → act as non-saturating current switch.

- The voltage-transfer characteristics is as follows:



(i) Region A: $V_{in} < V_R$

↳ Q1 OFF, Q2 ON (forward-active)

↳ $V_{o1} = V_{CC}$, $V_{o2} = V_{CC} - I_{EE}R_2$ [ideal case: $I_{E1} = 0$, $I_{E2} = I_{EE}$]

(ii) Region B: $V_{in} = V_R$

↳ Q1, Q2 ON (forward-active)

↳ $V_{o1} = V_{CC} - \left(\frac{I_{EE}}{2}\right)R_1$; $V_{o1} = V_{CC} - \left(\frac{I_{EE}}{2}\right)R_2$

(iii) Region C: $V_{in} > V_R$

↳ Q1 ON (forward-active), Q2 OFF

↳ $V_{o1} = V_{CC} - I_{EE}R_1$, $V_{o2} = V_{CC}$ [ideal case: $I_{E1} = I_{EE}$, $I_{E2} = 0$]

* $V_{out-} = V_{o1} - V_{BE(ON)}$ and $V_{out+} = V_{o2} - V_{BE(ON)}$

* V_R can be found using regular circuit analysis. To simplify calculations, we can assume base currents of transistors to be negligible, and all diodes / base-emitter junctions have $\Delta V = V_{BE(ON)}$ (typically, find the base voltage of the BJT stage before V_R, then use $V_E = V_B - V_{BE(ON)}$)

- The transition region is defined in terms of emitter current of Q_1/Q_2 as the change in i/p voltage when switching the emitter current from 5% to 95% of its final value.
- Note in practice, we cannot steer I_{EE} at. It only flows through one branch → $I_{EE} = I_{E1} + I_{E2}$.

$$\frac{I_{E1}}{I_{E2}} = \frac{I_{E0}\, e^{(V_{B1}-V_E)/V_t}}{I_{E0}\, e^{(V_R-V_E)/V_t}} = e^{(V_{B1}-V_R)/V_t} \rightarrow \boxed{V_{B1} = V_R - V_t \ln\left(\frac{I_E - I_{E1}}{I_{E1}}\right)}$$

At $I_{E1} = 0.05 I_E$, $V_{B1} = V_{IL} = V_R - V_t \ln\left(\frac{0.95}{0.05}\right)$; At $I_{E1} = 0.95 I_E$, $V_{B1} = V_{IH} = V_R - V_t \ln\left(\frac{0.05}{0.95}\right)$

$$\Delta V_i = V_{IH} - V_{IL} = 2V_t \ln 19 \approx 150mV, \text{ taking } V_t = 25mV.$$

- Typical values are $V_{EE} = -5.2V$, $V_R = -1.3V$, $I_{EE} = 2mA$, $V_{BE(on)} = 0.65V$

$V_{OH} = -0.9V$, $V_{OL} = -1.75V$, $V_{IH} = -1.225V$, $V_{IL} = -1.375$ → $NM_L = 0.375V$, $NM_H = 0.325V$, logic swing $= 0.85V$.

- ECL is the fastest logic circuit available conventionally. (BJT never in saturation → fast)
- Differential nature of circuit makes it resilient to noise and complementary o/ps are available.
- Current drawn from supplies remains const. during switching (unlike CMOS or TTL)
- Note the o/p voltage levels are not compatible w/ other logic families → hard to integrate.

Types of ECL

- Two of the most popular forms of commercially available ECL is ECL 10K and ECL 100K.
  - ↳ ECL 10K : lower power dissipation but is slow [gate delay = 2ns, power dissipation = 25mW]
  - ↳ ECL 100K: faster switching but higher power dissipation [gate delay = 0.75ns, power dissipation = 40mW]
- Other forms of ECL include PECL and LVPECL
  - ↳ Positive ECL (PECL) : $V_{CC} = 5V$, $V_{EE} = 0V$ → easier to interface w/ other logic families. (However, logic levels are defined rel. to supply rail, diff to other logic families)
  - ↳ Low Voltage PECL (LVPECL): $V_{CC} = 3.3V$, $V_{EE} = 0V$ → power optimised version of PECL.
- The voltage levels used in ECL, PECL and LVPECL are summarised below:

Type	V_{EE}	V_{OL}	V_{OH}	V_{CC}
ECL	-5.2V	-1.75V	-0.9V	GND
PECL	GND	3.4V	4.2V	5V
LVPECL	GND	1.0V	2.4V	3.3V

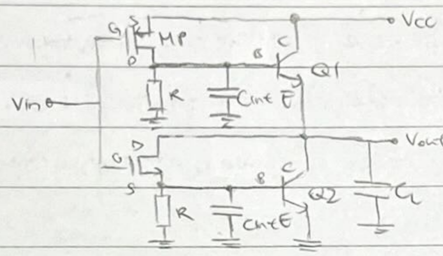
BiCMOS transistor and logic circuits

BiCMOS-transistor

- CMOS has low static power consumption, high i/p impedance and wide noise margins
  BJTs have a large current o/p → high fan-out
- We can amplify the o/p of MOSFET using BJTs to form a BiCMOS transistor.
  BiCMOS transistors have high i/p impedance, slightly worse noise margins but much better fan-out.

BiCMOS inverter

- The circuit diagram for the BiCMOS inverter is as follows:



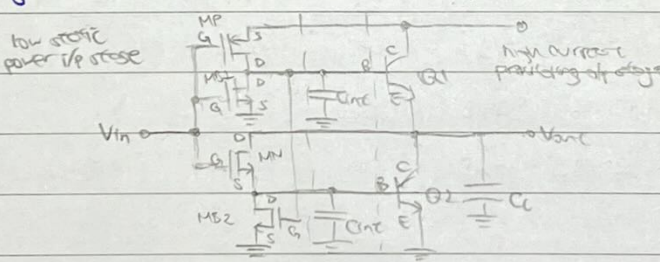
(i) $V_{in} > V_{IH}$ (Vin HIGH)
  ↳ MN is ON → base and collector of Q2 shorted
  ↳ Q2 starts conducting → Vout pulled to GND → C_L discharges through Q_2 ($V_{out} = V_{BE,2}$)

(ii) $V_{in} < V_{IL}$ (Vin LOW)
  ↳ MP is ON → base and collector of Q1 shorted
  ↳ Q1 starts conducting → Vout pulled to Vcc → C_L charges through Q1 ($V_{out} = V_{CC} - V_{BE,1}$)

* The base charges of Q1/Q2 (represented by Cint) discharges through the resistors.

BiCMOS inverter with active pull-up pull-down

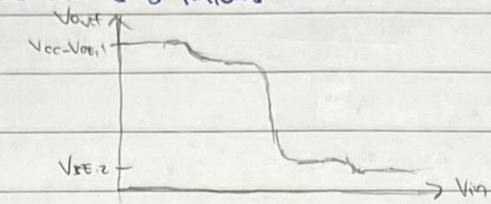
- The circuit diagram for the BiCMOS inverter w/ active pull-up pull-down is as follows:



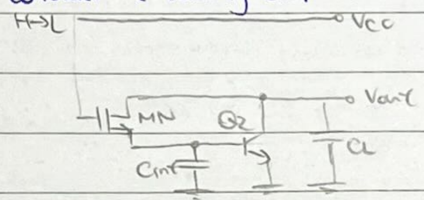
- Resistors increase power consumption and rq. large surface area → expensive. → Replace R w/ MB1, MB2
- For Vin LOW, MP ON, MB1 OFF / For Vin HIGH, MN ON, MB2 OFF → all current charge Cint
  When MP is OFF, MB1 is ON / when MN is OFF, MB2 is ON → Cint can discharge through MB1/MB2
* Operation essentially the same as regular BiCMOS inverter.
- Current o/ps of MOS transistors are low → use larger $\frac{W}{L}$ (or use cascaded stages w/ increasing $\frac{W}{L}$)
  Current is amplified at the o/p stage by BJTs (w/o increasing the overall static power consumption much)

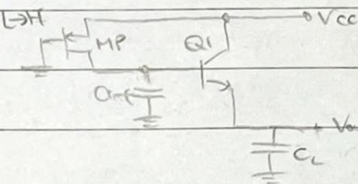
Performance of BiCMOS inverter

- The voltage transfer characteristic is as follows:



- Vout is upper bounded by $V_{CC}-V_{BE,1}$ and lower bounded by $V_{BE,2}$ → slightly worse V_{OH}, V_{OL} performance
  sharp fall → better inverter performance (good logic swing and noise margins)
- Both $V_{CE,1}$ and $V_{CE,2}$ are above saturation levels → Q1/Q2 never enter saturation → faster switching
- It is hard to fully analyse propagation delays in BiCMOS (many devices involved), so assume the following:
  ↳ i/p signal switching occurs instantaneously ↳ load is purely capacitive
  ↳ approximate MN, MP as resistors when conducting
- We will only consider the following simplified circuits





① H-L propagation delay t_{PHL}

↳ Once ON, replace MN w/ equivalent resistor R_N, $R_N = \frac{V_{CC}}{\frac{1}{2}k_N(V_{CC}-V_{TN})^2}$

↳ Propagation delay consists of (i) charging of C_{int} and (ii) Discharging of C_L

(i) C_{int} needs to be charged up to $V_{BE(on)}$ to make Q2 ON

$$t_1 = \frac{Q}{\langle I \rangle} = \frac{C_{int}V_{BE(on)}}{(V_{CC}-\frac{V_{BE(on)}}{2})/R_N} = \frac{C_{int}R_N V_{BE(on)}}{V_{CC}-0.5V_{BE(on)}}$$

(ii) Q2 ON → C_L discharged to $V_{CC}/2$ by Q2

capacitance referred from emitter to base

total capacitance at base side: $C_T = C_{int} + \frac{C_L}{\beta+1}$

$$t_2 = \frac{Q}{\langle I \rangle} = \frac{C_T V_{CC}/2}{V_{CC}/2R_N} = \frac{(C_{int}+C_L/\beta+1)V_{CC}/2}{V_{CC}/2R_N} = \left(C_{int}+\frac{C_L}{\beta+1}\right)R_N$$

$$\Rightarrow t_{PHL} = t_1+t_2 = \frac{C_{int}R_N V_{BE(on)}}{V_{CC}-0.5V_{BE(on)}} + \left(C_{int}+\frac{C_L}{\beta+1}\right)R_N$$

② L-H propagation delay t_{PLH}

↳ Once ON, replace MP w/ equivalent resistor R_P, $R_P = \frac{V_{CC}}{\frac{1}{2}k_P(V_{CC}-|V_{TP}|)^2}$

↳ Propagation delay consists of (i) charging of C_{int} and (ii) charging of C_L

(i) C_{int} needs to be charged up to $V_{BE(ON)}$ to make Q1 ON.

$$t_1 = \frac{Q}{\langle I \rangle} = \frac{C_{int}V_{BE(ON)}}{(V_{CC}-\frac{V_{BE(ON)}}{2})/R_P} = \frac{C_{int}R_P V_{BE(ON)}}{V_{CC}-0.5V_{BE(ON)}}$$

(ii) Q1 ON → C_L charged to $V_{CC}/2$ by Q1

total capacitance at base side: $C_T = C_{int} + \frac{C_L}{\beta+1}$

$$t_2 = \frac{Q}{\langle I \rangle} = \frac{C_T V_{CC}/2}{V_{CC}/2R_P} = \frac{(C_{int}+C_L/\beta+1)V_{CC}/2}{V_{CC}/2R_P} = \left(C_{int}+\frac{C_L}{\beta+1}\right)R_P$$

$$\Rightarrow t_{PLH} = t_1+t_2 = \frac{C_{int}R_P V_{BE(ON)}}{V_{CC}-0.5V_{BE(ON)}} + \left(C_{int}+\frac{C_L}{\beta+1}\right)R_P$$

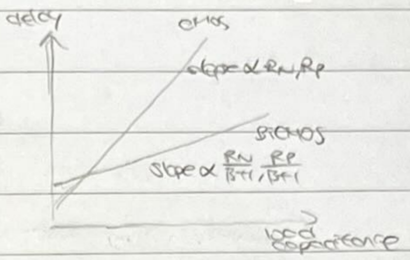
Comparison of CMOS and BiCMOS.

- For BiCMOS, $t_{PLH} = \frac{C_{int} R_P V_{BE(ON)}}{V_{CC} - 0.5 V_{BE(ON)}} + \left(C_{int} + \frac{C_L}{\beta+1}\right) R_P = aC_{int} + \frac{b}{\beta+1} C_L$, where $a = 1.2R_P$, $b = R_P$

  $t_{PHL} = \frac{C_{int} R_N V_{BE(ON)}}{V_{CC} - 0.5 V_{BE(ON)}} + \left(C_{int} + \frac{C_L}{\beta+1}\right) R_N = cC_{int} + \frac{d}{\beta+1} C_L$, where $c = 1.2R_N$, $d = R_N$

- For CMOS, $t_{PLH} = 0.7 R_P C_L$

  $t_{PHL} = 0.7 R_N C_L$



→ For low C_L, coeff a, c dominant → CMOS performs better

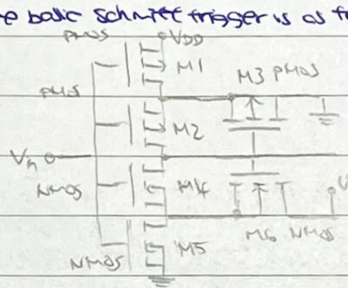
For high C_L, coeff b, d dominant → BiCMOS performs better

+ BiCMOS manufacturing process is more complex → greater cost per gate.

CMOS Schmitt trigger

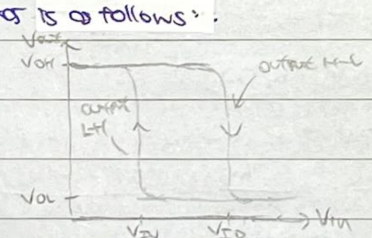
CMOS Schmitt trigger

- Schmitt triggers have hysteresis — H-L o/p and L-H o/p transitions occur at diff. i/p voltages.
- Schmitt triggers are useful for speeding up slowly changing signals and cleaning up noisy signals.
- The circuit diagram for the basic Schmitt trigger is as follows:



M3, M6 effectively FB signals from the o/p back to the series connected stack of transistors.

- The voltage transfer characteristics is as follows:



To determine the critical pts, assume that no current is drawn from the o/p, Vout.

(i) V_{OH} HIGH, assume $V_{in} = 0V$

↳ V_{in} at GND → M5 OFF → no pull down path to GND

↳ V_{in} at GND → M1 ON (non-saturation) → $V_{S,2}$ pulled to virtual V_{DD} → M2 ON (non-saturation)

↳ High conductance path btwn V_{DD} and V_{out} → $V_{OH} = V_{DD}$

(ii) V_{OL} LOW, assume $V_{in} = V_{DD}$

↳ V_{in} at V_{CC} → M1 OFF → no pull up path to V_{CC}

↳ V_{in} at V_{CC} → M5 ON (non-saturation) → $V_{S,4}$ pulled to virtual GND → M4 ON (non-saturation)

↳ High conductance path btwn GND and V_{out} → $V_{OL} = 0V$

→ logic swing = V_{DD} (good)

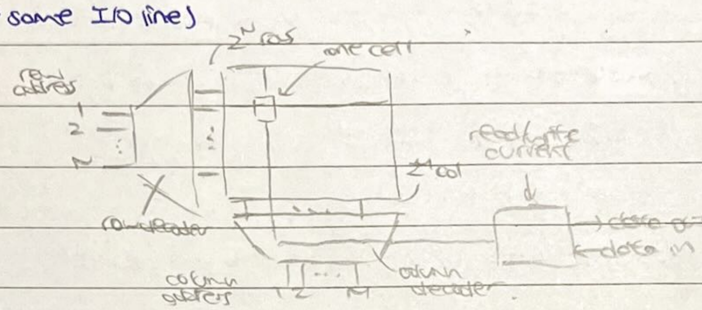
Semiconductor memories

Semiconductor memories

Memory elements

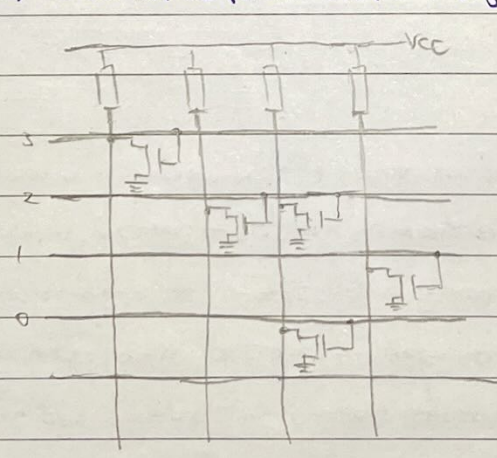
- A device having more than one bit of addressing req. addressing, where an address designates a specific storage location. The o/ps of the device are the values stored in these locations.
- There a three modules in a memory device.
  ↳ 1) Memory array — N×M array of one bit storage cells (total memory is $2^M \times 2^N = 2^{M+N}$)
  ↳ 2) Row decoder — For given N bit address i/p, it selects a unique horizontal line out of 2^N lines.
  (Each selection corresponds to a word, stored in the corresponding row of the memory array)
  ↳ 3) Column decoder — For given M bit i/p, selects from 2^M vertical lines for read/write operation.
  (Selects the bit inside the word selected by the row decoder).

* Only one row can be addressed at a time, or data becomes indeterminate (since cells on the same column share the same I/O line)



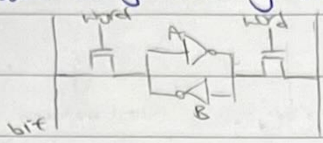
Read only memory (ROM)

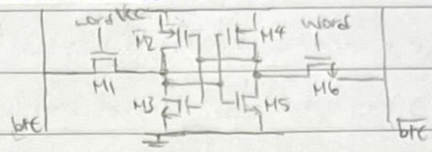
- A ROM's content is written only once (i.e. read only)
- Data stored is encoded by absence or presence of MOS transistors at row-column intersections.
  (Gate connected w/ address i/p, drain connected to columns, source grounded).
- Upon address selection, transistors on the corresponding line are turned ON
  ↳ If there is a transistor, it is turned ON → drain is pulled to GND → logic 0 is read
  ↳ If there is no transistor, the column line remains intact → logic 1 is read



Static random access memory (SRAM).

- For RAM, it is necessary to read out their data and also update/write the data.
- A basic RAM cell consists of a pair of cross-coupled MOS inverters. The inverters can be accessed through switching transistors, which are themselves connected to the row address lines.





- The RAM cell can be either logic 1 or logic 0.
  - ↳ Inverter B is generating logic 1/0 → inverter A o/ps logic 0/1 → inverter B o/ps logic 1/0. i.e. FB through the inverters, state is kept intact.
- To change the state of the RAM cell,
  - ↳ Drive the vertical bitlines to the new values
  - ↳ Operate the address lines to switch on the pass transistors — connect cell to bit lines.

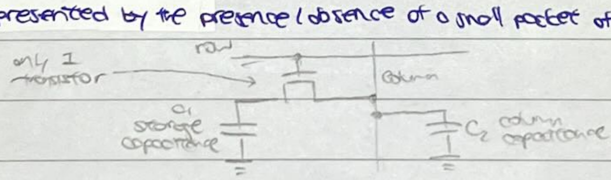
* The RAM cell inverters must be sufficiently "weak" s.t. the bit-line signals can override the current o/p values — use M2–M5 w/ smaller $\frac{W}{L}$, M1, M6 w/ larger $\frac{W}{L}$

- To read the state of the RAM cell,
  - ↳ Both bit lines are preset to precisely the same voltage level
  - ↳ Pass transistors activated → inverter o/ps perturb the bit lines by a few mV → eqm. disturbed (Extra circuitry added to speed up this process)
  - ↳ After reading, the pass transistors are turned OFF → RAM cell settles back to original value.

Dynamic random access memory (DRAM)

DRAM cheaper than SRAM when memory > 256kB.

- SRAM uses 6 transistors per bit and several control lines → high cost of memory per bit.
  DRAM solves this by using memory cells based on a capacitive storage element (reduced component count)
- Logic 0/1 is represented by the presence/absence of a small packet of charges stored in a capacitor.



- Leakage currents can remove the stored charge in a few ms → necessary to restore/refresh the logic signal more frequently than this → additional complexity in the control/drive circuitry.
- Read/write by applying a logic 1 to the gate of the transistor via the row
  - ↳ Data written into the cell by logic 1/0 on the column line while the cell is selected.
  - ↳ When reading, charges stored on C_1 is shared w/ much larger capacitance C_2 of the column line
    → we measure the change in potential (may be susceptible to noise → use differential sensing (+ve FB)

* $Q_{before} = C_1V_{c1} + C_2V_{c2}$, $Q_{after} = (C_1+C_2)(V_{c2}+\Delta V)$ → solve to find ΔV when logic 1/0 stored in C_1
we can solve for V_{c1} given a sensing threshold ΔV_{th} → find how freq. we need to refresh, $f = \frac{1}{t_{decay}}$
power dissipated found using $P = \frac{1}{2T_r}(V_i^2 - V_f^2)$, where $V_f = V_i - \frac{I_{leak}T_r}{C_1}$

(iii) Output H-L transition (V_{in} rises from GND)

↳ V_{in} rises above V_{TN} → M5 turns ON (saturation)

↳ $V_{G,6}$ still at $V_{out} = V_{OH} = V_{DD}$. since $V_{D,6} = V_{DD}$ → $V_{DS6} = V_{GS6} > V_{GS6} - V_{TN}$ → M6 saturated.

↳ while M4 remains non-conductive, equating drain currents I_{D5}, I_{D6}

$$I_{D,5(sat)} = I_{D,6(sat)} \rightarrow \frac{k_5}{2}(V_{in}-V_{TN})^2 = \frac{k_6}{2}(V_{GS,6}-V_{TN})^2 \rightarrow V_{GS,6} = \sqrt{\frac{k_5}{k_6}}(V_{in}-V_{TN}) + V_{TN}$$

↳ Noting that $V_{D,5} = V_{DD} - V_{GS,6}$, we have $V_{D,5} = V_{DD} - \sqrt{\frac{k_5}{k_6}}(V_{in}-V_{TN}) - V_{TN}$

↳ M4 first turns on when $V_{GS4} = V_{in} - V_{D,5} = V_{TN}$. At this pt, $V_{in} = V_{ID}$

$$V_{ID} = V_{D5} + V_{TN} = V_{DD} - \sqrt{\frac{k_5}{k_6}}(V_{ID} - V_{TN}) \rightarrow V_{ID} = \frac{V_{DD} + \sqrt{k_5/k_6}\,V_{TN}}{1 + \sqrt{k_5/k_6}}$$

(iv) Output L-H transition (V_{in} falls from V_{DD})

↳ V_{in} falls below $V_{DD} - |V_{TP}|$ → M1 turns ON (saturation)

↳ $V_{G,3}$ still at $V_{out} = V_{OL} = 0V$. since $V_{D,3} = 0V$ → $V_{DS,3} = V_{GS,3} > V_{GS,3} - |V_{TP}|$ → M3 saturated

↳ while M2 remains nonconductive, equating drain currents $I_{D,1}, I_{D3}$,

$$I_{D,1(sat)} = I_{D,3(sat)} \rightarrow \frac{k_1}{2}(V_{DD}-V_{in}-|V_{TP}|)^2 = \frac{k_3}{2}(|V_{GS,3}| - |V_{TP}|)^2 \rightarrow |V_{GS,3}| = \sqrt{\frac{k_1}{k_3}}(V_{DD}-V_{in}-|V_{TP}|) + |V_{TP}|$$

↳ Noting that $V_{D,1} = 0 + |V_{GS,3}|$, we have $V_{D,1} = \sqrt{\frac{k_1}{k_3}}(V_{DD}-V_{in}-|V_{TP}|) + |V_{TP}|$

↳ M2 first turns on when $|V_{GS,2}| = V_{D,1} - V_{in} = |V_{TP}|$. At this pt, $V_{in} = V_{IU}$

$$V_{IU} = V_{D1} - |V_{TP}| = \sqrt{\frac{k_1}{k_3}}(V_{DD} - V_{IU} - |V_{TP}|) \rightarrow V_{IU} = \frac{\sqrt{k_1/k_3}\,(V_{DD}-|V_{TP}|)}{1+\sqrt{k_1/k_3}}$$

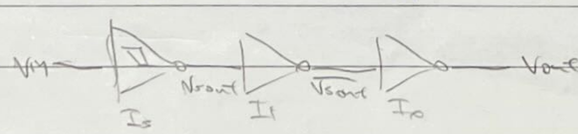
* V_{ID} is the o/p H-L trip voltage and V_{IU} is the o/p L-H trip voltage, $V_{IL} > V_{IU}$

CMOS Schmitt trigger device dimensions

- V_{ID} and V_{IU} depend on the process gain factor k of the FB devices (M3, M6), which in turn depend on the $(\frac{W}{L})$ of the corresponding transistors.
- Increasing the $(\frac{W}{L})$ ratio for M3, M6 results in greater diff. btwn V_{ID}, V_{IU} and the mid-pt voltage $\frac{V_{DD}}{2}$, giving greater hysteresis, and vice versa

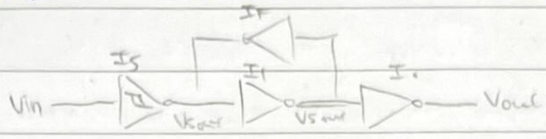
Buffered Schmitt trigger

- The i/p stage of CMOS Schmitt inverter consists of 4 MOS transistors in series. For comparable currents w/ CMOS inverter (2 MOS transistors), each device must have double the $(\frac{W}{L})$ ratio
- The increase in size of the devices result in
  ↳ increase the i/p capacitance to an unchargeable amount
  ↳ decrease in resistance $R \sim 1/(W/L)$
- If we req. a CMOS Schmitt inverter w/ high drive capability is req. it is best to use the smallest possible devices in the i/p stage (min. capacitance), followed by a two-stage inverting buffer.



Schmitt trigger with regenerative feedback.

- A FB inverter I_F can improve the transient response of a CMOS Schmitt trigger w/ noisy i/p.



- I_F provides further noise immunity by applying +ve FB. → o/p switching more sharply/positively.
- To avoid contention, forward inverter I_1 should have more drive capability than I_F.

(we can achieve this by using transistors of lower k for I_F)

Combinational and sequential circuits

Combinational circuits

Boolean algebra

- The theorems of switching algebra are as follows:

Commutation	$a+b=b+a$	$ab=ba$
Association	$(a+b)+c=a+(b+c)$	$(ab)c=a(bc)$
Distribution	$a(b+c)=ab+ac$	$a+(bc)=(a+b)(a+c)$
Absorption	$a+ab=a$	$a(a+b)=a$
De-Morgan	$\overline{a+b}=\bar{a}\bar{b}$	$(a+b)(a+\bar{b})=a$

SOP, POS and canonical form

- The SOP is made of minterms (m) and the POS is made of maxterms (M).
  There are 2^n minterms and maxterms (n is the no. of variables).
  ↳ e.g. For variables a,b,c ($n=3$), $m_0=\bar{a}\bar{b}\bar{c}$, $m_1=\bar{a}\bar{b}c$ ... $m_7=abc$
  $M_0=a+b+c$, $M_1=a+b+\bar{c}$ ... $M_7=\bar{a}+\bar{b}+\bar{c}$
- In canonical form, all the binary variables appear once in each term (SOP) or factor (POS)
  To expand an expression into its canonical form, we use $x+\bar{x}=1$ → we get BCD $f=\Sigma(\ )$.
  or get via K-map
- For SOP, expand the BCD $f=\Sigma(\ )$
  For POS, find BCD for $\bar{f}$, $\bar{f}=\Sigma(\ )$, expand, then apply De-Morgan's.
* Canonical form is useful for MUX, ROM, LUT; minimised form is useful w/ using min. no. of logic gates.

Karnaugh maps.

- Karnaugh maps for 3/4/5 variables are shown below.

0	2	6	4
1	3	7	5

abc

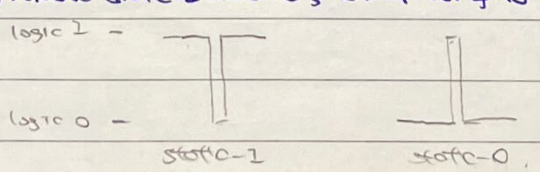
0	4	12	8
1	5	13	9
3	7	15	11
2	6	14	10

abcd

0	8	24	16	1	9	25	17
2	10	26	18	3	11	27	19
6	14	30	22	7	15	31	23
4	12	28	20	5	13	29	21

abcde

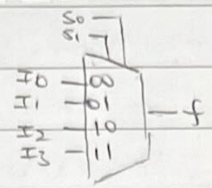
- We can easily find the BCD of a function f or its inversion $\bar{f}$ using a K-map.
  (use $f=\Sigma(\ )$ for SOP/NAND implementation, $\bar{f}=\Sigma(\ )$ for POS (NOR implementation))
- We can remove static hazards by ensuring all product terms overlap in the K-map.
  (K-map for f to remove static-1 hazard; K-map for $\bar{f}$ to remove static-0 hazard)



Multiplexers (MUX)

- A MUX uses the select i/ps S to select which data i/p I to output.

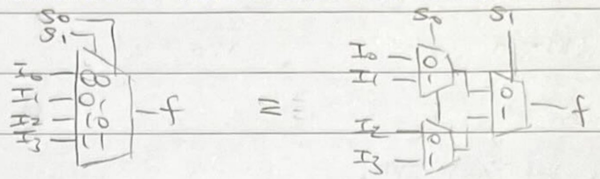
↳ e.g. For a 4-to-1 MUX,



S_1	S_0	f
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

SOP implementation: $f = \bar{S}_1\bar{S}_0 I_0 + \bar{S}_1 S_0 I_1 + S_1\bar{S}_0 I_2 + S_1 S_0 I_3$

- Larger MUXs can be constructed from smaller ones (smallest is 2-to-1 MUX)



- MUX can be used to realise combinational circuits

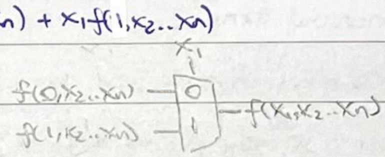
↳ using ext. i/ps only as select i/ps → LUT.

↳ using some ext. i/ps as select i/ps → Shannon's expansion thm.

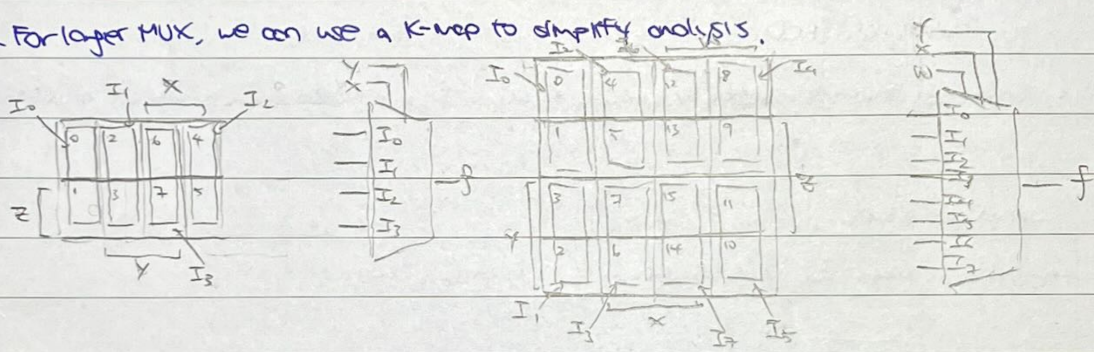
- Shannon's expansion thm. states any boolean function $f(x_1,x_2..x_n)$ can be written in the form

$$f(x_1,x_2..x_n) = \bar{x}_1 f(0,x_2..x_n) + x_1 f(1,x_2..x_n)$$

We can use a 2-in-1 MUX w/ x_1 as select i/p.



- For larger MUX, we can use a K-map to simplify analysis.



The data i/ps are functions of z. This can be found using a truth table

I=0

z	I
0	0
1	0

I=z

z	I
0	0
1	1

I=$\bar{z}$

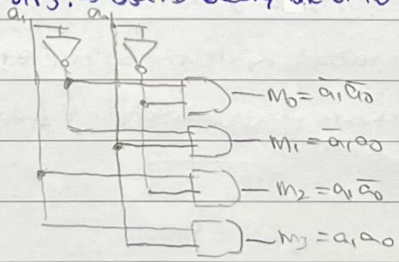
z	I
0	1
1	0

I=1

z	I
0	1
1	1

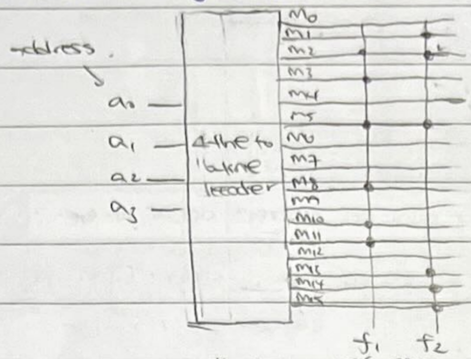
Decoders

- A decoder has n i/ps and 2^n o/ps. → asserts exactly one of its o/ps depending on the i/p combination.



Read only memory (ROM)

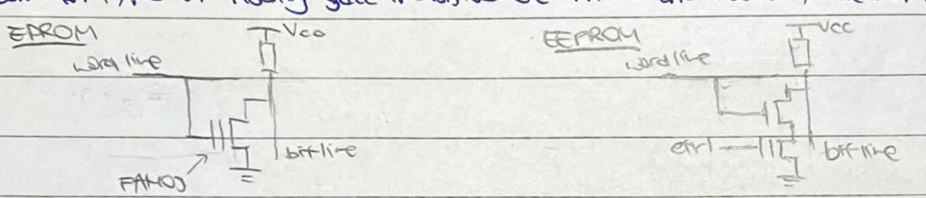
- ROM is an interconnected array of semiconductor devices to store an array of binary data. stored data cannot be changed under normal operation conditions.



16 word x 2 bit ROM.

$f_1 = \Sigma(2,3,5,8,10,11)$
$f_2 = \Sigma(1,2,5,13,14,15)$

- • in the ROM matrix indicates a programmable link — equivalent to storing a "1".
- Each cell in the ROM is implemented using floating gate transistors. (FAMOS / FLOTOX).
  - ↳ Floating gate transistors are programmed "1" by applying a high voltage → es trapped on the floating gate → V_{th} shifts → transistor never conducts during normal operation (1)
  - ↳ If the floating gate transistor is not programmed "0", it acts like a regular NMOS transistor → word line high → transistor conducts, providing a path to GND (0).
- common types of floating gate transistors are FAMOS and FLOTOX, used in EPROM and EEPROM respectively.

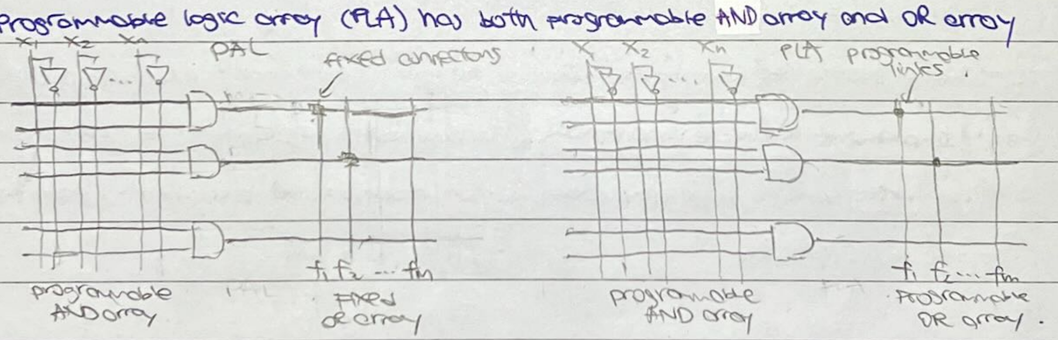


To program a "1", EPROM: Gate and drain ~20V ; EEPROM: Drain Low, Ctrl ~12V

To erase "1", EPROM: UV light ; EEPROM: Drain ~12V, Ctrl Low.

Programmable logic

- Programmable array logic (PAL) has a programmable AND array and fixed OR array. Programmable logic array (PLA) has both programmable AND array and OR array



* ROM has a fixed AND array and a programmable OR array.

Sequential logic

Bistable circuits

- The excitation table for bistable circuits is as follows:

output $Q \to Q^+$	Required state S R		J K		T	D
$0 \to 0$	0	x	0	x	0	0
$0 \to 1$	1	0	1	x	1	1
$1 \to 0$	0	1	x	1	1	0
$1 \to 1$	x	0	x	0	0	1

* For SR bistable, changes occur at the rising clock edge ($S=1, R=1$ is not allowed)

For JK bistable, changes occur at the falling clock edge ($J=1, K=1$ toggles the o/p)

Models for sequential circuits.

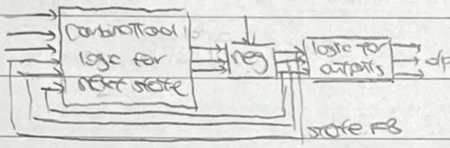
- For a Moore machine, the o/ps are a function of the current state, $Z=f(Q)$, $Q^+=f(x,Q)$

  the o/p changes synchronously w/ state changes.

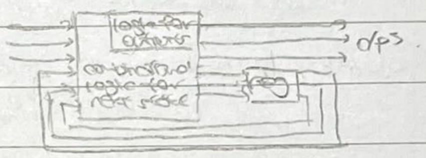
- For a Mealy machine, the o/ps depend on state and on i/ps, $Z=f(x,Q)$, $Q^+=f(x,Q)$

  i/p changes can cause immediate o/p changes (asynchronous)

Moore machine



Mealy machine.



Design of synchronous circuits

- Sequential circuits use a clock to synchronise all bistables' operation. The design steps are:
  - ↳ 1) Design the problem and draw the state diagram
  - ↳ 2) Draw the state table, remove redundant states and allocate bistables
  - ↳ 3) Draw the state transition table (using excitation table for bistable circuits)
  - ↳ 4) Draw a K-map for each bistable and the o/p.
  - ↳ 5) Draw the circuit diagram
- To merge states, the next state must be the same / don't care for each i/p column.

  However, they need not have the same o/p (though extra logic is req — Mealy machine)

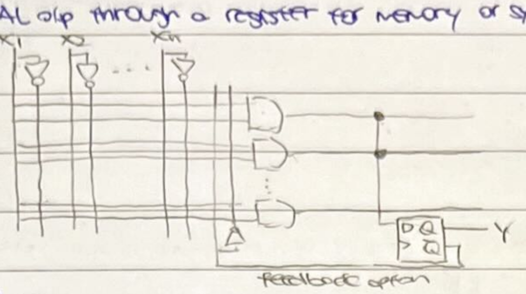
- If we order the i/ps using gray code, we can read-off the K-map entries from the state transition table.

Programmable logic circuits, data storage processing and control

Programmable logic circuits.

Sequential PAL.

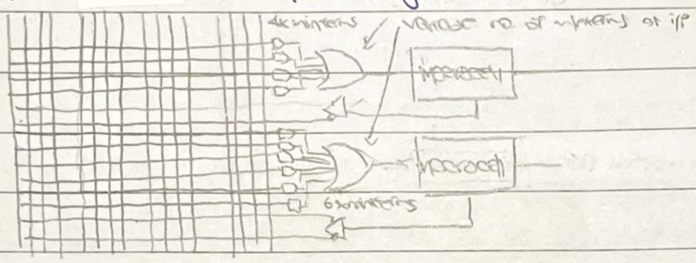
- We can feed the PAL o/p through a register for memory or synchronisation — sequential PAL



- Advantages of PAL:
  - ↳ Fewer devices req. → smaller footprint, lower cost, power savings, easier to test + debug.
  - ↳ High design flexibility, in system reprogrammability (same technology as EEPROM)

Programmable logic device (PLD)

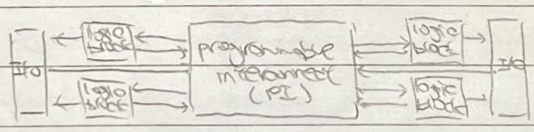
- A PLD is formed by an multiple PAL arrays in a single device



- Each macrocell is programmable and its function can vary depending on the MUX select i/ps. It can switch btwn combinational/sequential circuits or i/p/o/p select.

Complex PLD (CPLD)

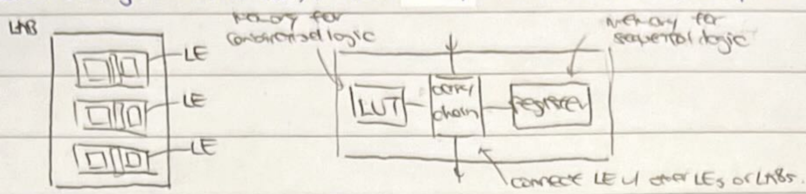
- A CPLD is formed by combining multiple logic array blocks (LAB) in a single device w/ programmable interconnect and I/O (control & pins)



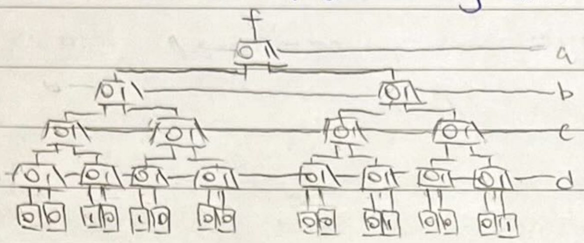
- Each LAB is effectively a PLD, but instead of variable no. of minterms as OR gate i/p, makes use of local programmable interconnect and expander product terms.
- Note that the expander product terms have an extra delay, but this is deterministic and can be accounted for in timing simulations.
- The PI is a global routing that connects any signal to any destination in device (programmed w/ EEPROM)
- The I/O control blocks are separated from LAB by PI. Tri-state buffer control is used to enable i/ps, o/ps or bidirectional on any I/O pin. (In PLD, I/O connected directly to logic)
- Advantages of CPLD:
  - ↳ High flexibility in logic and routing (greater capability than PLDs) reprogrammable.
  - ↳ Instantly turn on at board power up (same for PLDs)

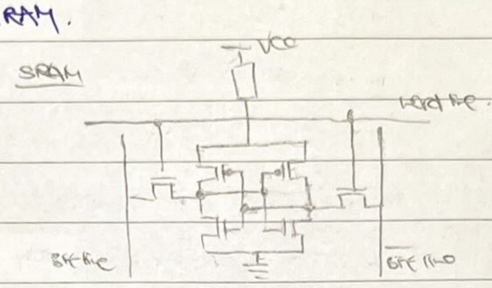
Field Programmable Gate Arrays (FPGA)

- FPGAs consist of LABs (based on logic elements), memory blocks, DSP blocks, PLLs and transceivers.
- Each LAB is made of logic elements (LE). A LE mainly consists of a LUT, carry chain and register.



- A lookup table (LUT) replaces the product term array and is using memory as a combinational circuit.
- A LUT is formed using MUXs (where ext. i/ps are only used as select i/ps) and its contents (i.e. functionality) are programmed using EEPROM or SRAM.

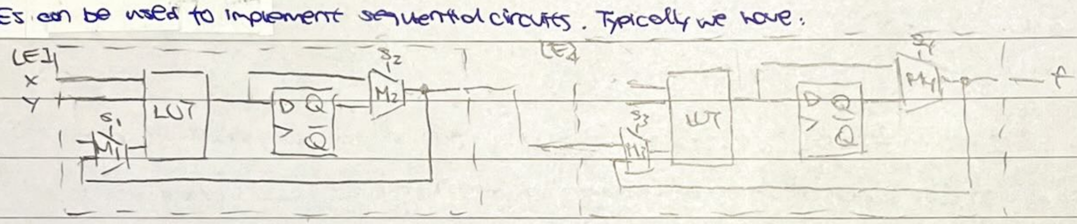




LUT = 0010 1000 0001 0001 → $f = \Sigma(2, 4, 11, 15)$

similar to a bistable to store a bit

- SRAM is volatile – req. power to maintain the stored information
- LEs can be used to implement sequential circuits. Typically we have:



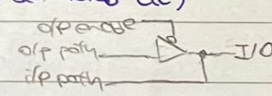
$S_1 S_2 S_3 S_4 = 1100$ → LE1 as sequential circuit, LE2 as combinational circuit

* LE2 LUT has ext. i/p X, Y → Mealy machine ; LE2 LUT no ext. i/p X, Y → Moore machine.

- In a FPGA, the LABs are arranged in an array, connected w/ interconnects
  - ↳ local interconnect: connects btwn LEs within a LAB / direct connections btwn adjacent LABs
  - ↳ row and col interconnect: fixed length routing segments that span a no. of LABs / entire device.
- The I/O elements in FPGAs control I/O features (i/p vs o/p vs bidirectional, i/o standards etc)

  ↙ controls tri-state buffer.

  A typical I/O element consists of o/p enable control, o/p path and i/p path



- FPGAs use PLLs to generate clocks and DLLs to lock the phase quickly. The clock control blocks enable/disable clocks and control the clock routing network (which span the entire device).

  In a lab, we can program using JTAG

- Most FPGAs use SRAM to program PI and LUT → use ext. EEPROM, CPLD or CPU to program (Active: FPGA controls program sequence automatically ON ; Passive: CPU controls program sequence).
- Advantages of FPGA:
  - ↳ high performance (high density to create complex logic performance), integration of many functions
  - ↳ fast programming, low cost (compared to ASIC for low quantities).

Data storage, processing and control

Data storage.

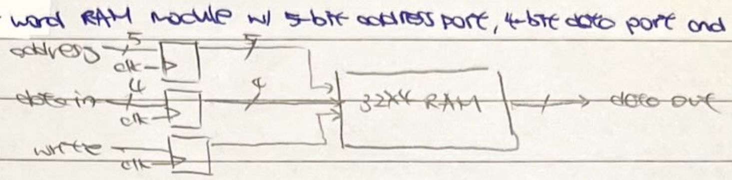
- The main types of memory are ROM and RAM.
  - ↳ Read-only memory (ROM): read only, not suitable as working space.
  - ↳ Random access memory (RAM): read and write, much faster.
- Both ROM and RAM are random access — all addresses are accessible in an equal amount of time and can be selected in any order.
- FPGAs have dedicated memory blocks that can be constructed into ROM or single/dual port RAM

  one address for read/write

  one address for read, one address for write

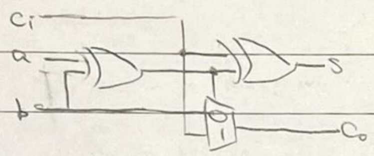
  some memory blocks are memory LABs (MLAB) — LABs configured as memory.

  ↳ e.g: 32 4-bit word RAM module w/ 5-bit address port, 4-bit data port and a write control i/p.



Arithmetic circuits.

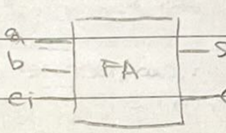
- A full-adder (FA) circuit works as follows



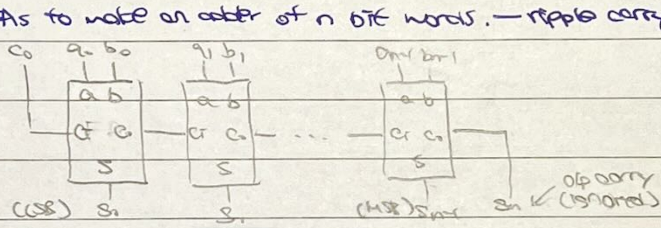
$$s = a \oplus b \oplus c$$

$$C_o = \overline{(a \oplus b)}b + (a \oplus b)c$$

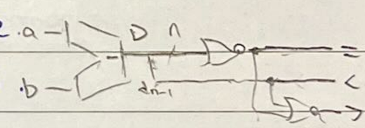
$$= ab + ac + bc$$



- We can cascade n FAs to make an adder of n bit words. — ripple carry adder circuit

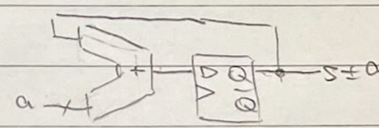


- We can invert one of the i/ps and set $C_0 = 1$ to add the 2's complement → subtraction.
- We can make a comparator based on subtractors
  - ↳ $a = b$ when $a - b = 0$ → check o/p is zero by using n-bit NOR gate.
  - ↳ $a < b$ when $a - b < 0$ → check MSB (-ve number has 1 as MSB)
  - ↳ $a > b$ when $a - b$ is not $=0$ or <0 → pass $=, <$ o/p into NOR gate

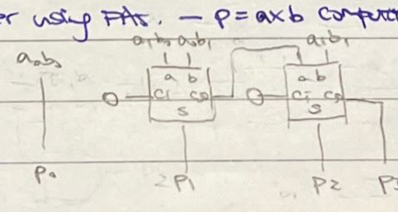


- We can make an accumulator using an adder and a register

  (counters are accumulators w/ $a = 1$ always)



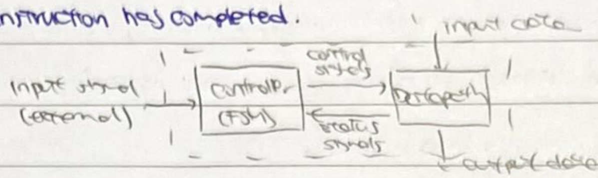
- We can make a multiplier using FAs. — $P = a \times b$ computed as an addition of summands



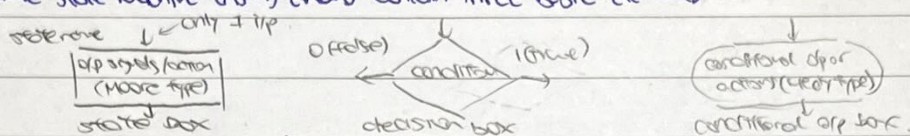
$$\begin{array}{rrrr} & & a_1 & a_0 \\ \times & & b_1 & b_0 \\ \hline & & a_1b_0 & a_0b_0 \\ & a_1b_1 & a_0b_1 & \\ \hline & P_2 & P_1 & P_0 \end{array}$$

Digital systems

- A digital system consists of a control unit FSM (processor) that controls registers, MUXs and adder/subtractors. It executes operations specified in the form of instructions.
- The FSM steps through the instructions, asserting the control signals needed in successive clock cycles until the instruction has completed.



- The control circuit implements an algorithm — finite set of instructions which terminate in a finite time at a known end state. This has many representations — pseudocode, ASM charts etc.
- Algorithmic state machine (ASM) charts contain three basic elements:



← for each state.

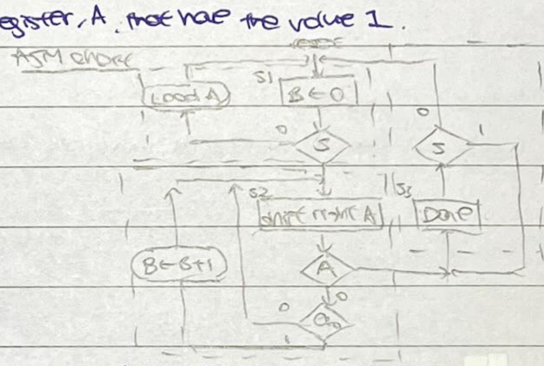
- In ASM charts, the entire block is considered as one unit — all operations within block occur during single edge transition (whereas in a conventional flowchart, one action follows another)
- The next state is evaluated during the same clock. System then enters the next state during transition of the next clock.

- e.g.: we wish to count the no of bits in a register, A, that have the value 1.

pseudocode

```
B=0;
while A≠0 do
  if a0=1 then
    B=B+1;
  end if;
  right-shift A;
end while
```

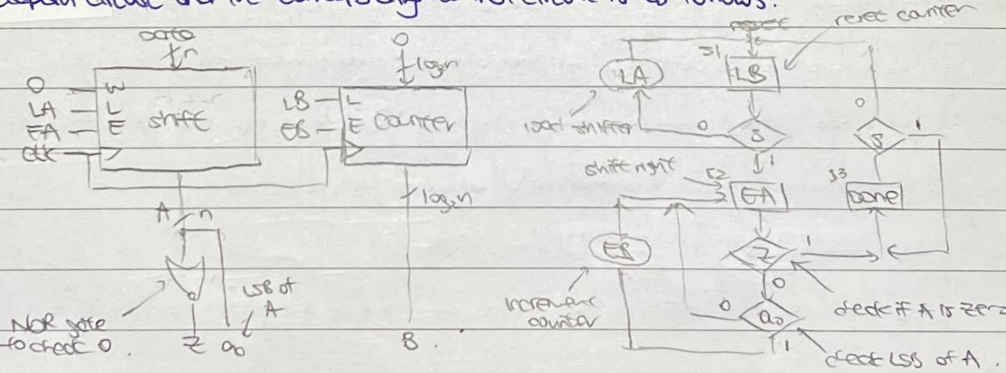
ASM chart



↳ Enter state → evaluate next state → transition to next state + Moore/Mealy type o/p or action during the rising edge of the next clock cycle.

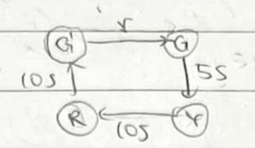
← block diagram

↳ The datapath circuit and the corresponding control circuit is as follows:



- eg: Implement a simple synchronous circuit for the following traffic light controller.
  - Initial state vehicle GREEN
  - After a pedestrian request is made, (i) change state vehicle YELLOW for 5 seconds, (ii) change state vehicle RED for 10 seconds (iii) change state vehicle GREEN' for 10 seconds, (iv) return to initial state.

① State diagram



② State table + bistable allocation

Present state		Next state (r, 5s, 10s)								output		
state	Q_1Q_2	000	001	011	010	110	111	101	100	G	Y	R
G	0 0					Y	Y	Y	Y	1	0	0
Y	0 1			R	R	R	R			0	1	0
R	1 1		G'	G'			G'	G'		0	0	1
G'	1 0		G	G			G	G		1	0	0

doesn't depend on r, 5s, 10s → moore machine.

③ State transition table

present state		Next state ($Q_1^+Q_2^+$)								Bistable inputs (D_1D_2)								output		
state	Q_1Q_2	000	001	011	010	110	111	101	100	000	001	011	010	110	111	101	100	G	Y	R
G	0 0	00	00	00	00	01	01	01	01	00	00	00	00	01	01	01	01	1	0	0
Y	0 1	01	01	11	11	11	11	01	01	01	01	11	11	11	11	01	01	0	1	0
R	1 1	11	10	10	11	11	10	10	11	11	10	10	11	11	10	10	11	0	0	1
G'	1 0	10	00	00	10	10	00	00	10	10	00	00	10	10	00	00	10	1	0	0

④ Karnaugh maps

D_1 (r; Q_2; Q_1; 10s; 10s; 5s)

0	0	0	0	0	0	0	0
0	0	1	1	1	1	0	0
1	1	1	1	1	1	1	1
1	0	0	1	1	0	0	1

D_2 (r; Q_2; Q_1; 10s; 10s; 5s)

0	0	0	0	1	1	1	1
1	1	1	1	1	1	1	1
1	0	0	1	1	0	0	1
0	0	0	0	0	0	0	0

$$D_1 = Q_1Q_2 + Q_1\overline{10s} + 5sQ_2$$

$$D_2 = \overline{Q_1}Q_2 + r\overline{Q_1} + \overline{10s}Q_2$$

G (Q_1; Q_2)

1	1
0	0

$G = \overline{Q_2}$

Y (Q_1; Q_2)

0	0
1	0

$Y = \overline{Q_1}Q_2$

R (Q_1; Q_2)

0	0
0	1

$R = Q_1Q_2$

⑤ Circuit diagram

This can be implemented in many ways: logic gates, PLA, PAL, ROM, FPGA etc.

VHDL

VHDL basics

VHDL design unit

- VHDL code for a high-level component has the following structure.

```
entity high_level_component is
    generic declaration
    port declaration
end high_level_component;
architecture
    component(s) declaration
    type declaration
    constant declaration
    signal declaration
begin
    component instantiation – generic map, port map
end behaviour;
```

- VHDL code for a low-level component has the following structure.

```
entity low_level_component is
    generic declaration
    port declaration
end low_level_component;
architecture
    type declaration
    constant declaration
    signal declaration
begin
    concurrent blocks
    sequential blocks (process)
end
```

- sometimes it may be useful use libraries/modules

```
library IEEE;
use ieee.std_logic_1164.all;
```

① Generic declaration — parameterises the entity

↳ e.g: `generic(n: NATURAL := 4; K: INTEGER);`

② Port declaration — describes i/ps and o/ps

↳ e.g:
```
port(x1, x2 : IN BIT;
          f : OUT BIT);
```

③ Type declaration — defines custom datatypes (useful for states)

↳ e.g: `type state_type is (s1, s2, s3);`

④ Constant declaration — defines constants

↳ e.g: `constant K : INTEGER := 3;`

⑤ Signal declaration — defines signals (connections)

↳ e.g: `signal y : state_type;`

⑥ Component declaration — includes low-level component in high-level component

↳ e.g:
```
component counter is
    generic(n: NATURAL := 4);
    port(clock : IN STD_LOGIC; count : OUT STD_LOGIC_VECTOR(n downto 0));
end component;
```

⑦ Component instantiation — configures the low level component

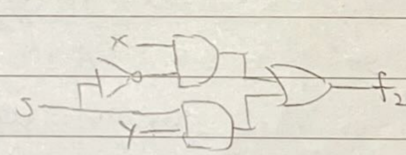
↳ e.g:
```
slow_clock : counter
    generic map (n => 27);
    port map (clock => CLOCK_50, count => BCD0);
```

Concurrent assignment statements

- The order in which they appear in VHDL code does not affect the meaning of the code (represent implied processes that execute in parallel)

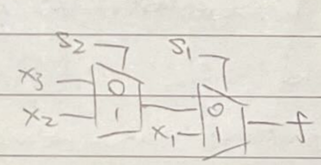
(i) Simple signal assignment

↳ e.g: `f2 <= (NOT s AND x) OR (s AND y);`



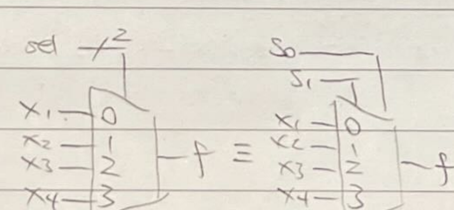
(ii) Conditional signal assignment (2-to-1-MUX)

↳ e.g:
```
f <= x1 when s1 = '1' else
     x2 when s2 = '1' else x3;
```



(iii) Selected signal assignment (general MUX)

↳ e.g:
```
with sel select
    f <= x1 when "00",
         x2 when "01",
         x3 when "10",
         x4 when OTHERS;
```



sequential assignment statements

- The ordering of the statements may affect the meaning of the code (placed inside a process statement — explicit processes).
- A process has the following structure :

```
process_name : process ( sensitivity_list )
    type declaration
    constant declaration
    variable declaration
begin
    sequential statement
end process
```

① Variable declaration — defines variables (local variables)

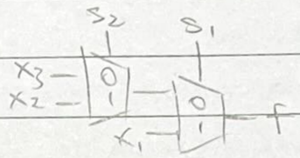
↳ e.g: `variable count : INTEGER := '0';`

② Sequential statement — indicate behaviour and express order

(i) if-then statements

↳ e.g:

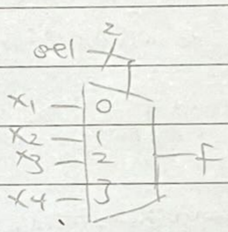
```
if s1='1' then f <= x1;
elsif s2='2' then f <= x2;
else f <= x3;
end if;
```



(ii) case statements (general MUX)

↳ e.g:

```
case sel is
    when "00" => f <= x1;
    when "01" => f <= x2;
    when "10" => f <= x3;
    when OTHERS => f <= x4;
end case;
```



signal and variable assignment

- signals are global and can be assigned values using the <= operator
- variables are local (to processes) and can be assigned values using the := operator

std_logic_vector data type.

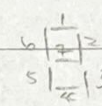
- The std_logic_vector data type is part of the ieee.std_logic_1164 module.
- A std_logic_vector can be defined using TO or DOWNTO.
  ↳ e.g.: LED : OUT std_logic_vector(0 TO 3)

LSB	f_0	f_1	f_2	f_3	MSB
	LED(0)	LED(1)	LED(2)	LED(3)	

LED : OUT std_logic_vector(3 DOWNTO 0)

MSB	f_3	f_2	f_1	f_0	LSB
	LED(3)	LED(2)	LED(1)	LED(0)	

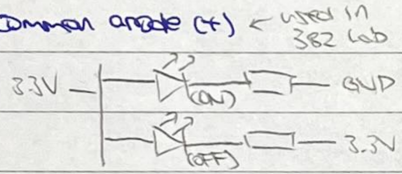
- Using DOWNTO is generally more natural, but we use TO by convention in certain applications (e.g. 7-segment display)
- We can slice a std_logic_vector in the following ways:
  ↳ All bits: LED <= "1010"; ↳ Single bit LED(0) <= '0';
  ↳ slicing: LED(3 DOWNTO 2) <= "10"; or LED(2 TO 3) <= "01";



7-segment display

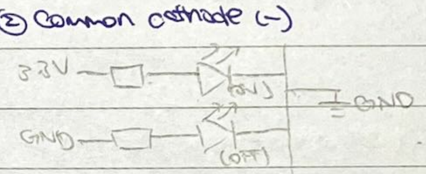
- 7-segment displays display decimal numbers. LED segments are connected to common pins (+/-)

① Common anode (+) ← used in 3B2 Lab.



set '0' to turn on LED

② Common cathode (-)



set '1' to turn on LED

In general, we can consider a controlled inverter → change s as required.

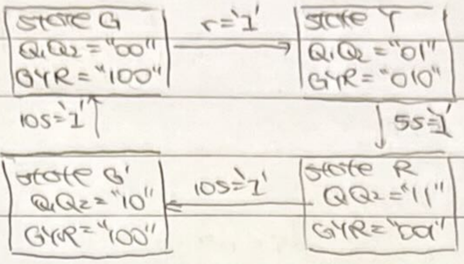
common anode $s=1$, $a=\overline{a'}$
common cathode $s=0$, $a=a'$

- We can implement a 7-segment decoder by considering the truth table → K-map → logic gate / PLA / PAL / ROM / LUT implementation

* The truth table for common anode and common cathode are inverse of each other.

VHDL for sequential circuits.

- When designing a Moore machine, we can separate the FSM and the o/ps into two processes.
- e.g. consider the following state diagram




```
entity tlc is
    port(clk : IN STD_LOGIC;
         r, 5s, 10s : IN STD_LOGIC;
         G, Y, R : OUT STD_LOGIC);
end tlc;
architecture behaviour of tlc is
    type state_type is (G, Y, R, G');
    signal state : state_type;
begin
FSM : process(clk)
begin
    if rising_edge(clk) then
        case state is
            when G => if r='1' then state <= Y;
                      else state <= G; end if;
            when Y => if 5s='1' then state <= R;
                      else state <= Y; end if;
            when R => if 10s='1' then state <= G';
                      else state <= R; end if;
            when G' => if 10s='1' then state <= G;
                      else state <= G'; end if;
        end case;
    end if;
end process;
```

alternatively use "wait until rising_edge(clk);"
or "if (clk'event AND clk='1') then"

```
outputs : process(state)
begin
    case state is
        when G OR G' => G='1';
        when Y => Y='1';
        when OTHERS => R='1';
    end case;
end process
end behaviour;
```